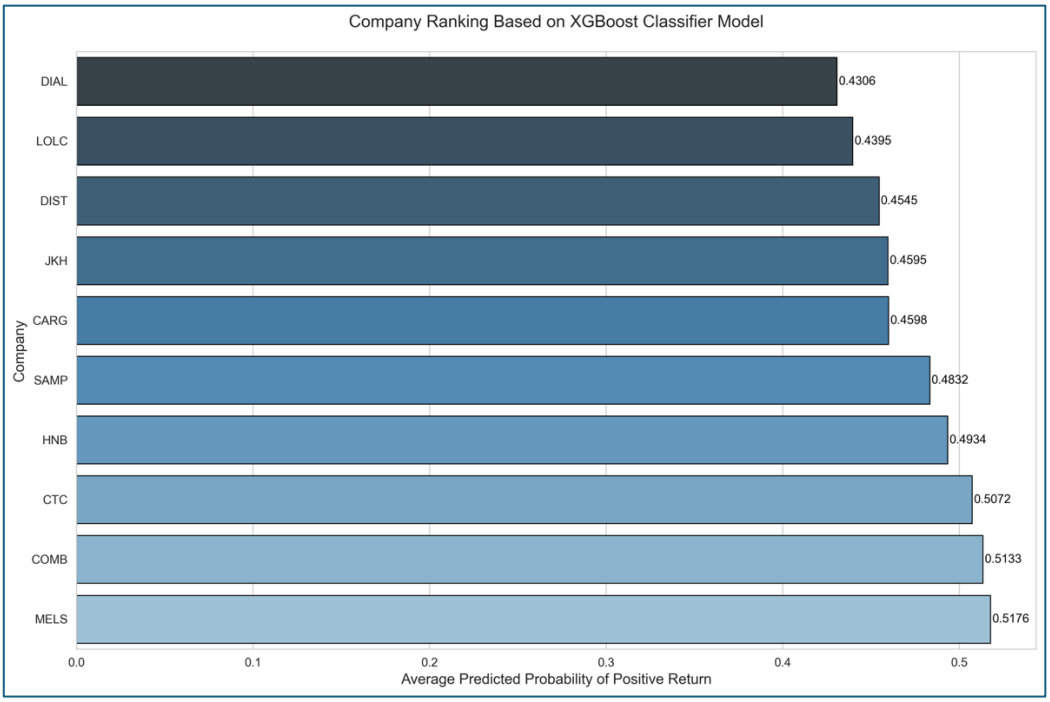
Classisfication Models:

- Logistic Regression AUC: 0.7846 ± 0.0684
- Random Forest AUC: 0.7912 ± 0.0593
- XGBoost AUC: 0.7933 ± 0.0521

Company	XGB_Pred_Prob	Rank
MELS	0.5176	1
COMB	0.5133	2
CTC	0.5072	3
HNB	0.4934	4
SAMP	0.4832	5
CARG	0.4598	6
JKH	0.4595	7
DIST	0.4545	8
LOLC	0.4395	9
DIAL	0.4306	10

Rank 1 = company with the highest average predicted probability of high future return according to the XGBoost classifier.



- XGB Classifier Rank IC: 0.7411
- XGB Classifier IC-IR: 28.8658
- XGB Classifier Top-3 Hit Rate: 0.292

Regression Models Accuracy

- Random Forest Regression R^2: 0.2832 ± 0.1383
- XGBoost Regression R^2: 0.2311 ± 0.1123

Ranking Models

❖ LightGBM Ranker

Company	LightGBM_Score	Overall_Rank
DIAL	-0.6908	1
CARG	-0.6924	2
MELS	-0.7191	3
DIST	-0.8018	4
CTC	-0.8708	5
COMB	-0.897	6
HNB	-0.9112	7
SAMP	-0.9287	8
JKH	-0.9993	9
LOLC	-1.0297	10

- LightGBM Rank IC: 0.7492
- LightGBM IC-IR: 4.8445
- LightGBM Top-3 Hit Rate: 0.803

❖ XGBoost Ranker

Company	XGBr_Score	Overall_Rank
DIAL	0.0733	1
HNB	0.0720	2
COMB	0.0609	3
SAMP	0.0590	4
MELS	0.0510	5
CARG	0.0415	6
CTC	0.0339	7
DIST	0.0304	8
JKH	-0.0028	9
LOLC	-0.0221	10

- XGBoost Rank IC: 0.7844
- XGBoost IC-IR: 4.1291
- XGBoost Top-3 Hit Rate: 0.739

LightGBM & XGBoost Rankers

Company	LightGBM_Rank	XGBoost_Rank
DIAL	1	1
CARG	2	6
MELS	3	5
DIST	4	8
CTC	5	7
COMB	6	3
HNB	7	2
SAMP	8	4
JKH	9	9
LOLC	10	10

Three Models Ranks

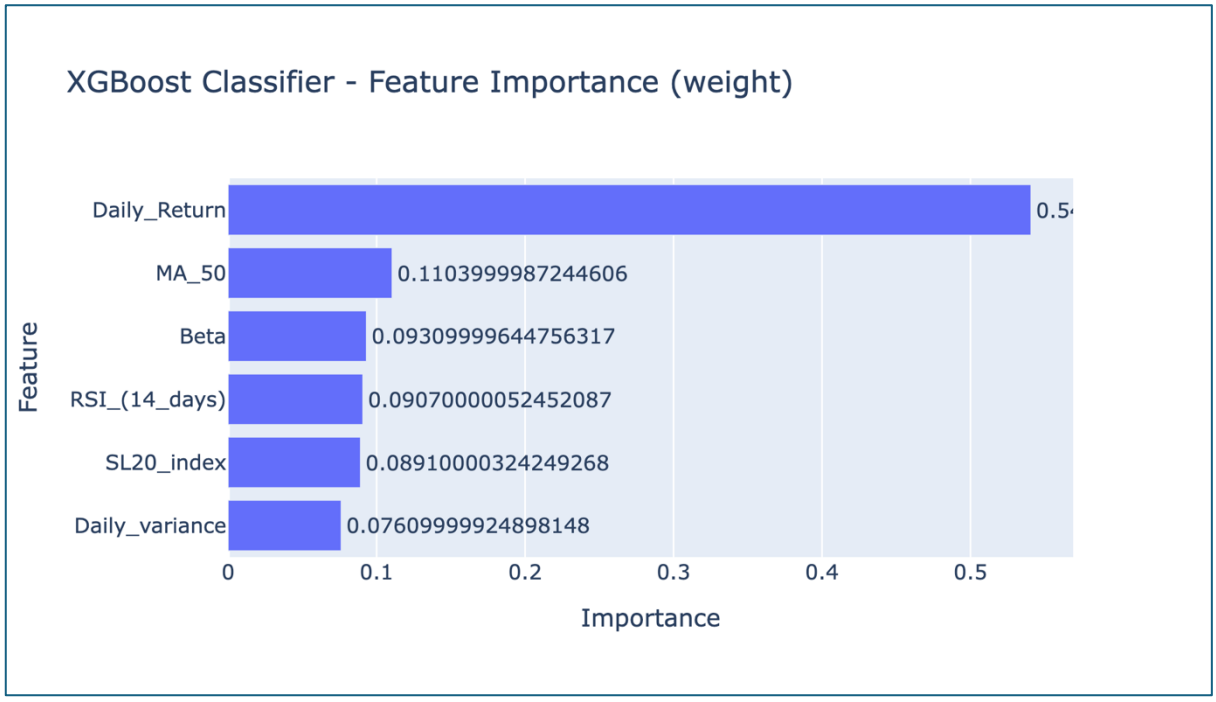
Company	XGB Classifier Score	XGB Classifier Rank	LightGBM Score	LightGBM Rank	XGBoost Ranker Score	XGBoost Ranker Rank
MELS	0.5176	1	-0.7191	3	0.0510	5
COMB	0.5133	2	-0.8970	6	0.0609	3
DIAL	0.4306	10	-0.6908	1	0.0733	1
HNB	0.4934	4	-0.9112	7	0.0720	2
CARG	0.4598	6	-0.6924	2	0.0415	6
CTC	0.5072	3	-0.8708	5	0.0339	7
SAMP	0.4832	5	-0.9287	8	0.0590	4
DIST	0.4545	8	-0.8018	4	0.0304	8
JKH	0.4595	7	-0.9993	9	-0.0028	9
LOLC	0.4395	9	-1.0297	10	-0.0221	10

Feature Importance

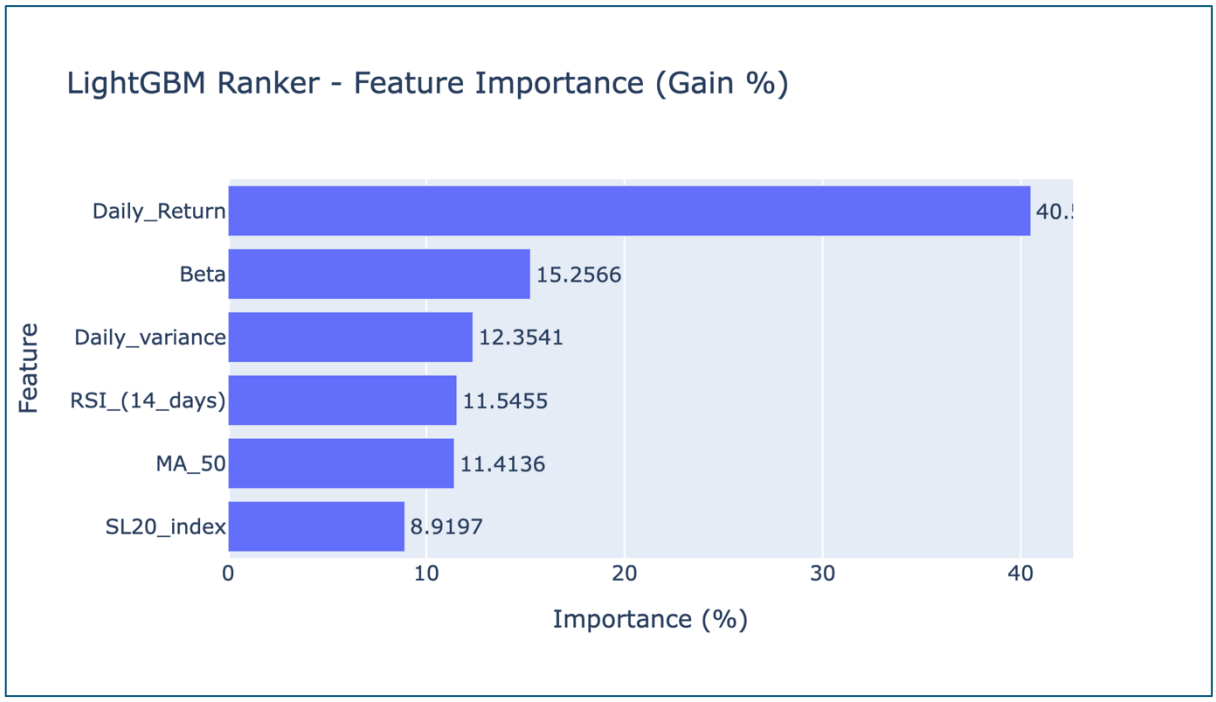
Combined Feature Importance Across All Three Models

Feature	XGB Classifier Weight	LightGBM Gain (%)	XGBoost Ranker Weight
Daily_Return	0.5407	40.5104%	0.2965
Beta	0.0931	15.2566%	0.1551
MA_50	0.1104	11.4136%	0.1412
RSI_(14_days)	0.0907	11.5455%	0.1372
Daily_variance	0.0761	12.3541%	0.1363
SL20_index	0.0891	8.9197%	0.1338

XGB Classifier Feature Importance



LightGBM Ranker Feature Importance



XGB Ranker Feature Importance

